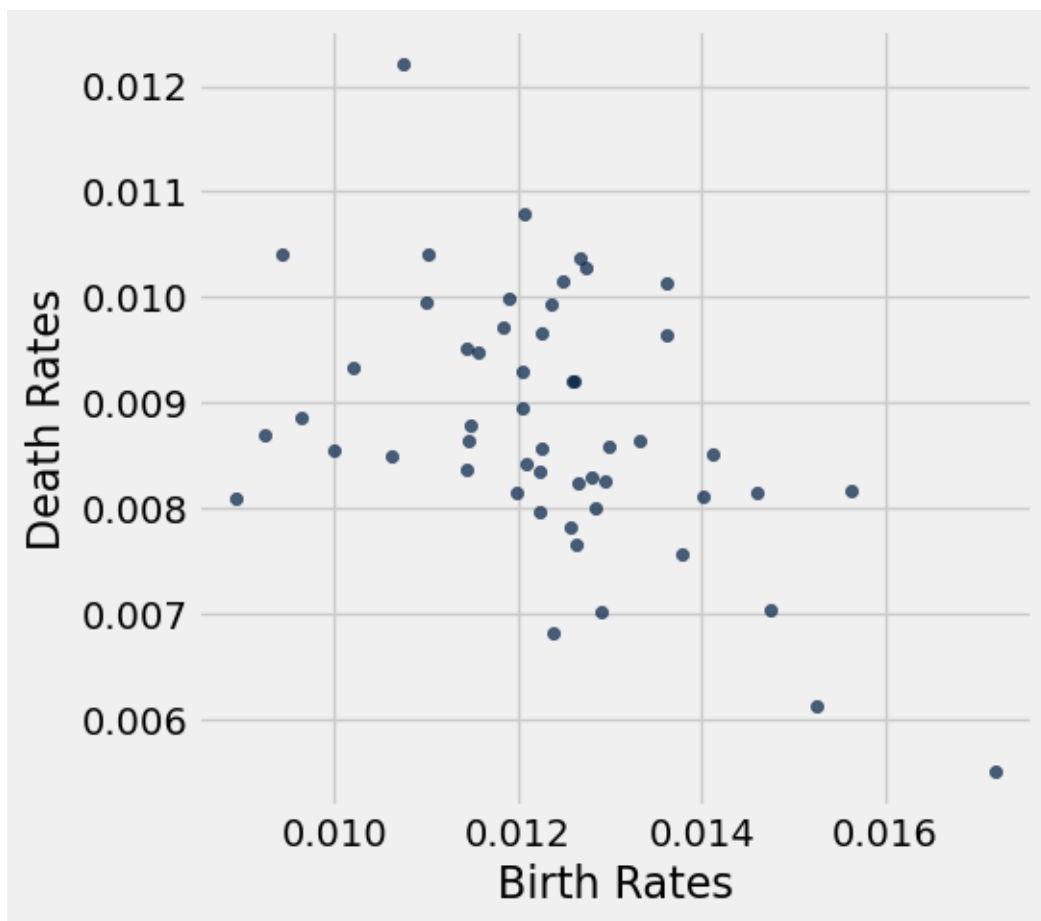


Question 5. In the code cell below, create a visualization that will help us determine if there is an association between birth rate and death rate during this time interval. It may be helpful to create an intermediate table containing the birth and death rates for each state. **(4 Points)**

Things to consider:

- What type of chart will help us illustrate an association between 2 variables?
- How can you manipulate a certain table to help generate your chart?
- Check out the [Recommended Reading](#) for this homework!

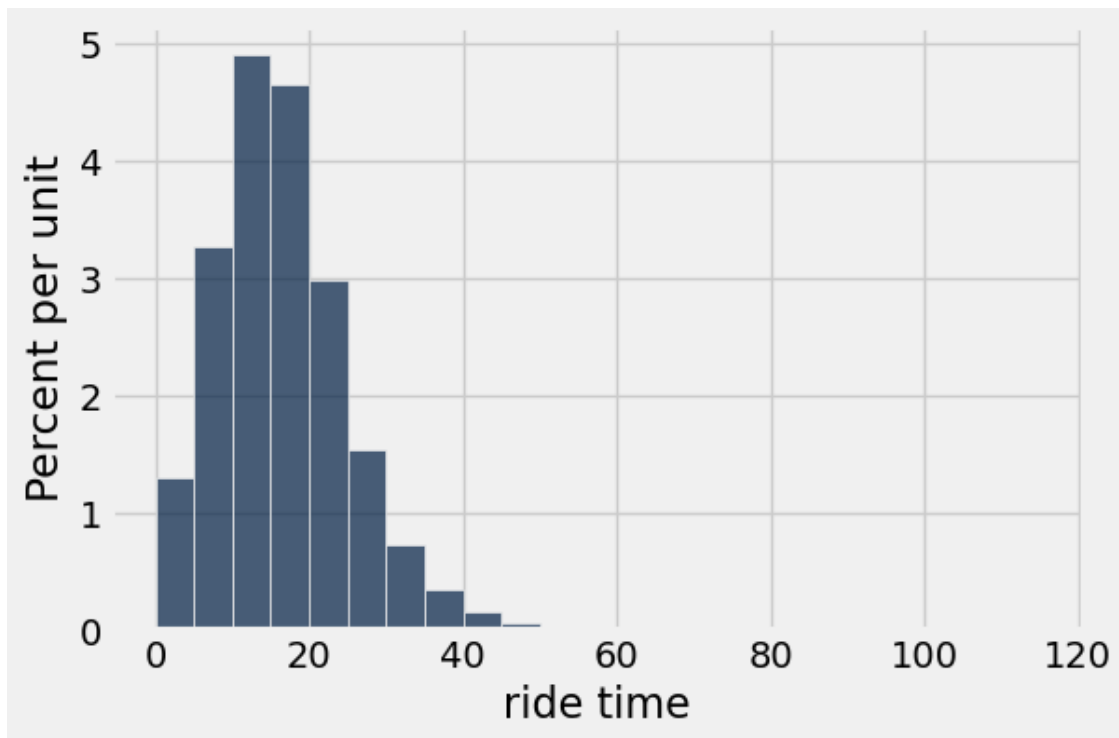
```
In [30]: # In this cell, use birth_rates and death_rates to generate your visualization
birth_rates_2015 = pop.column('BIRTHS') / pop.column('2015')
death_rates_2015 = pop.column('DEATHS') / pop.column('2015')
rates = Table().with_columns('Birth Rates', birth_rates_2015,
                             'Death Rates', death_rates_2015)
rates.scatter('Birth Rates', 'Death Rates')
```



Question 1. Produce a histogram that visualizes the distributions of all ride times in Boston using the given bins in `equal_bins`. (4 Points)

Hint: See [Chapter 7.2](#) if you're stuck on how to specify bins.

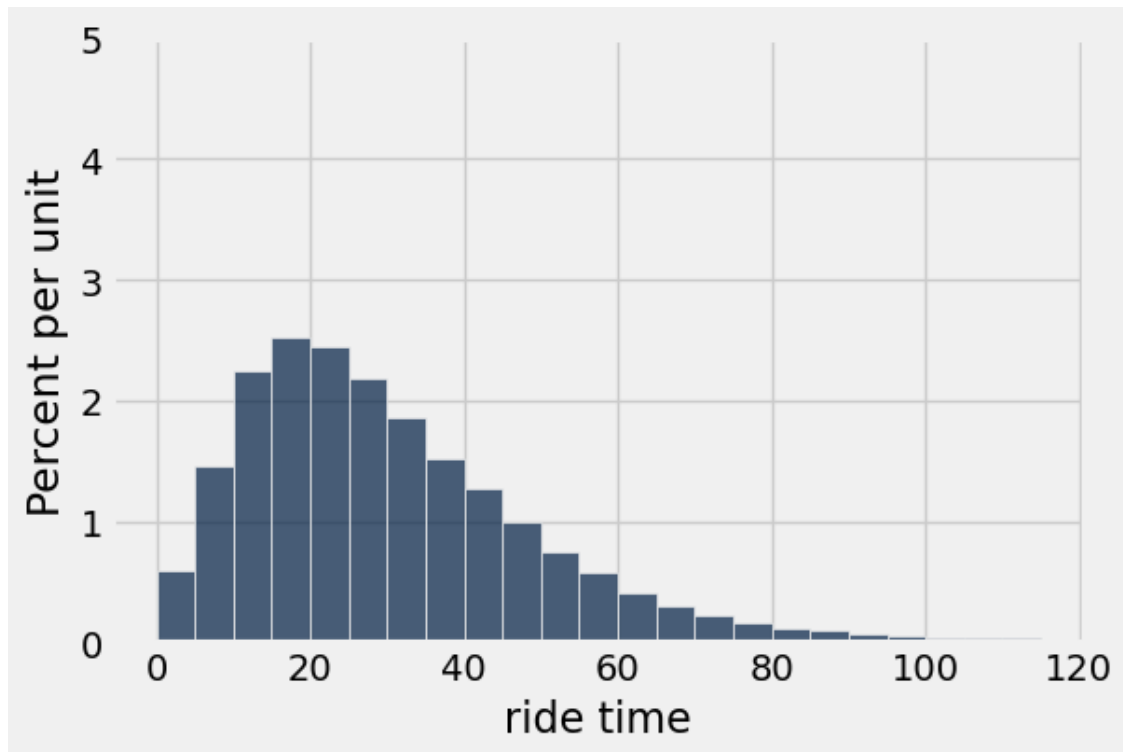
```
In [34]: equal_bins = np.arange(0, 120, 5)
        boston.hist('ride time', bins = equal_bins)
```



Question 2. Now, produce a histogram that visualizes the distribution of all ride times in Manila using the given bins. (4 Points)

```
In [35]: equal_bins = np.arange(0, 120, 5)
         manila.hist('ride time', bins = equal_bins)

         # Don't delete the following line!
         plt.ylim(0, 0.05);
```



Question 6. Identify one difference between the histograms, in terms of the statistical properties. > *Hint:* Without performing any calculations, can you comment on the average or skew of each histogram? (4 Points)

The manila histogram appears to be less evenly distributed than the boston histogram, meaning that it is more skewed to the right than the boston histogram. The manila histogram also appears to have a greater average ride time than the boston histogram because it has a greater range and no major outliers.

Question 7. Why is your solution in Question 6 the case? Based on one of the following two readings, why are the distributions for Boston and Manila different? **(4 Points)**

- [Boston reading](#)
- [Manila reading](#)

Hint: Try thinking about external factors of the two cities that may be causing the difference! The readings provide some potential factors – try to connect them to the ride time data.

Traffic laws are seldom enforced in Manila, spreading plenty of bottlenecks out across numerous roads for dozens of miles, in turn leading to slowdowns. The U-turns in Manila are disruptive since the roads are inefficiently designed, causing many drivers to navigate around the U-turns in a slow and cautious manner, which also leads to backups and higher ride times. Additionally, buses in Boston generally travel shorter distances within the city, which can contribute to the much shorter ride times in Boston.

Question 2. State at least one reason why you chose the histogram from Question 1. **Make sure to clearly indicate which histogram you selected** (ex: “I chose histogram A because ...”). **(5 Points)**

The scatter plot shows that plots are pointed within all values for x from -3 to 3, meaning that there should be no gaps in the histogram. Additionally, most of the points are plotted between $x = -1$ and $x = 0$, meaning that the highest percent per unit on the histogram for the x -values should be between $x = -1$ and $x = 0$. I chose histogram C because it shows that the highest percent per unit is between $x = -1$ and $x = 0$.

Question 4. State at least one reason why you chose the histogram from Question 3. **Make sure to clearly indicate which histogram you selected** (ex: “I chose histogram A because ...”). **(5 Points)**

The scatter plot shows a gap between $y = -0.5$ and $y = 0.5$, meaning that no plots are pointed within that range. I chose histogram B because there is no data between -0.5 and 0.5 in histogram B.

